

Q1.

Alex and Sam are playing a zero-sum game.

The game is represented by the pay-off matrix for Alex.

		Sam		
Alex	Strategy	S₁	S₂	S₃
	A₁	2	2	3
	A₂	0	3	5
	A₃	-1	2	-2

- (a) Explain why the value of the game is 2

(3)

- (b) Identify the play-safe strategy for each player.

(1)

(Total 4 marks)

Q2.

Victoria and Albert play a zero-sum game. The game is represented by the following pay-off matrix for Victoria.

		Albert			
		Strategy	X	Y	Z
Victoria	P	3	-1	1	
	Q	-2	0	1	
	R	4	-1	-1	

- (a) Find the play-safe strategies for each player.

(3)

- (b) State, with a reason, the strategy that Albert should never play.

(1)

- (c) (i) Determine an optimal mixed strategy for Victoria.

(5)

- (ii) Find the value of the game for Victoria.

(1)

- (iii) State an assumption that must be made in order that your answer for part

(c)(ii) is the maximum expected pay-off that Victoria can achieve.

(1)

(Total 11 marks)

Q3.

John and Danielle play a zero-sum game which does not have a stable solution.

The game is represented by the following pay-off matrix for John.

		Danielle			
		Strategy	X	Y	Z
John	A	2	1	−1	
	B	−3	−2	2	
	C	−3	−4	1	

Find the optimal mixed strategy for John.

(Total 6 marks)

Q4.

Harry and Will play a zero-sum game. The game is represented by the following pay-off matrix for Harry.

		Will			
Harry	Strategy	D	E	F	G
	A	4	-1	2	3
	B	4	6	3	7
	C	1	3	-2	4

(a) Show that this game has a stable solution and state the play-safe strategy for each player.

(4)

(b) List any saddle points.

(1)

(Total 5 marks)

Q5.

Romeo and Juliet play a zero-sum game. The game is represented by the following pay-off matrix for Romeo.

Juliet			
Strategy	D	E	F
A	4	-4	0



Romeo

B	-2	-5	3
C	2	1	-2

- (a) Find the play-safe strategy for each player. (3)
- (b) Show that there is no stable solution. (1)
- (c) Explain why Juliet should never play strategy D. (1)
- (d) (i) Explain why the following is a suitable pay-off matrix **for Juliet**.

4	5	-1
0	-3	2

- (ii) Hence find the optimal strategy for Juliet. (7)
- (iii) Find the value of the game for Juliet. (1)

(Total 15 marks)

Q6.

- (a) Two people, Adam and Bill, play a zero-sum game. The game is represented by the following pay-off matrix for Adam.

		Bill		
		B₁	B₂	B₃
Adam	A₁	-6	-1	-5
	A₂	5	2	-3
	A₃	-5	4	-4
	A₄	2	1	-4

- (i) Show that this game has a stable solution. (3)
- (ii) Find the play-safe strategy for each player. (1)
- (iii) State the value of the game for **Bill**. (1)
- (b) Roza plays a different zero-sum game against a computer. The game is represented by the following pay-off matrix for Roza.

Roza	Strategy	C₁	C₂	C₃
	R₁	3	4	-3
	R₂	-2	-1	5

- (i) State which strategy the computer should never play, giving a reason for your answer. (1)
- (ii) Roza chooses strategy R₁ with probability p . Find expressions for the expected gains for Roza when the computer chooses each of its two remaining strategies. (2)
- (iii) Hence find the value of p for which Roza will maximise her expected gains. (2)
- (iv) Find the value of the game for Roza. (1)

(Total 11 marks)

Q7.

- (a) Two people, Tom and Jerry, play a zero-sum game. The game is represented by the following pay-off matrix for Tom.

		Jerry		
Tom	Strategy	A	B	C
	I	-4	5	-3
	II	-3	-2	8
	III	-7	6	-2

Show that this game has a stable solution and state the play-safe strategy for each player.

(4)

- (b) Rohan and Carla play a different zero-sum game for which there is no stable solution. The game is represented by the following pay-off matrix for Rohan.

		Carla		
Rohan	Strategy	C₁	C₂	C₃
	R₁	3	5	-1
	R₂	1	-2	4

- (i) Find the optimal mixed strategy for Rohan and show that the value of the game is $\frac{3}{2}$.

(7)

- (ii) Carla plays strategy C_1 with probability p , and strategy C_2 with probability q .

Find the values of p and q and hence find the optimal mixed strategy for Carla.

(4)

(Total 15 marks)

Q8.

Two people, Roseanne and Collette, play a zero-sum game. The game is represented by the following pay-off matrix for Roseanne.

		Collette		
		Strategy	C ₁	C ₂
Roseanne	R ₁	−3	2	3
	R ₂	2	−1	−4

- (a) (i) Find the optimal mixed strategy for Roseanne.

(7)

- (ii) Show that the value of the game is -0.5 .

(1)

- (b) (i) Collette plays strategy C_1 with probability p and strategy C_2 with probability q . Write down, in terms of p and q , the probability that she plays strategy C_3 .

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(1)

- (ii) Hence, given that the value of the game is -0.5 , find the optimal mixed strategy for Collette.

(4)

(Total 13 marks)

Q9.

Two people, Rose and Callum, play a zero-sum game. The game is represented by the following pay-off matrix for Rose.

		Callum		
		C_1	C_2	C_3
Rose	R_1	5	2	−1
	R_2	−3	−1	5



R_3	4	1	-2
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- (a) (i) State the play-safe strategy for Rose and give a reason for your answer. (2)
- (ii) Show that there is no stable solution for this game. (2)
- (b) Explain why Rose should never play strategy R_3 . (1)
- (c) Rose adopts a mixed strategy, choosing R_1 with probability p and R_2 with probability $1 - p$.
- (i) Find expressions for the expected gain for Rose when Callum chooses each of his three possible strategies. Simplify your expressions. (3)
- (ii) Illustrate graphically these expected gains for $0 \leq p \leq 1$. (2)
- (iii) Hence determine the optimal mixed strategy for Rose. (3)
- (iv) Find the value of the game. (1)
- (Total 14 marks)

Q10.

Sam is playing a computer game in which he is trying to drive a car in different road conditions. He chooses a car and the computer decides the road conditions. The points scored by Sam are shown in the table.

		Road Conditions		
		C_1	C_2	C_3
Sam's Car	S_1	-2	2	4
	S_2	2	4	5
	S_3	5	1	2

Sam is trying to maximise his total points and the computer is trying to stop him.

- (a) Explain why Sam should never choose S_1 and why the computer should not choose C_3 . (2)
- (b) Find the play-safe strategies for the reduced 2 by 2 game for Sam and the computer, and hence show that this game does not have a stable solution. (4)

- (c) Sam uses random numbers to choose S_2 with probability p and S_3 with probability $1 - p$.
- (i) Find expressions for the expected gain for Sam when the computer chooses each of its two remaining strategies. (3)
- (ii) Calculate the value of p for Sam to maximise his total points. (2)
- (iii) Hence find the expected points gain for Sam. (1)
- (Total 12 marks)**

Q11.

Two people, Rowan and Colleen, play a zero-sum game. The game is represented by the following pay-off matrix for Rowan.

		Colleen			
		Strategy	C_1	C_2	C_3
Rowan	R_1		-3	-4	1
	R_2		1	5	-1
	R_3		-2	-3	4

- (a) Explain the meaning of the term 'zero-sum game'. (1)
- (b) Show that this game has no stable solution. (3)
- (c) Explain why Rowan should never play strategy R_1 . (1)
- (d) (i) Find the optimal mixed strategy for Rowan. (7)
- (ii) Find the value of the game. (1)
- (Total 13 marks)**

Q12.

- (a) Afzal and Bill play a zero-sum game. The game is represented by the following pay-off matrix for Afzal.

		Bill			
		I	II	III	IV



Afzal	I	3	-2	1	2
	II	3	5	2	6
	III	0	2	-3	3

(i) Show that this game has a stable solution.

(3)

(ii) List any saddle points.

(1)

(b) Colin and David play a zero-sum game. The game is represented by the following pay-off matrix for Colin.

		David	
		I	II
Colin	I	$x + 2$	$x - 1$
	II	3	5

Given that the optimal mixed strategy makes the value of the game $\frac{19}{5}$, find the value of x .

(8)

(Total 12 marks)

Q13.

Two people, Ann and Bev, play a zero sum game. The game is represented by the following pay-off matrix for Ann.

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		Bev		
		I	II	III
Ann	Strategy			
	I	1	3	6
	II	6	1	1
	III	7	4	2

(a) Show that there is no stable solution.

(3)

(b) Explain why it will never be optimal for Ann to adopt **Strategy II** and write down the reduced pay-off matrix for Ann.

(2)

(c) Hence, given that the value of the game is $3\frac{3}{5}$, find the optimal mixed strategy for

Bev.

(You are **not** required to find the optimal mixed strategy for Ann.)

(5)

(Total 10 marks)

Q14.

Alan and Tracy play squash against each other every week. They each have three shots that they can play to try to win a point: a boast (B), a drop shot (D) and a lob (L). The pay-off matrix for Alan is given below.

		Tracy		
		B	D	L
Alan	B	12	5	2
	D	5	-2	-3
	L	8	10	13

- (a) State which shot Alan should **not** play, giving a reason for your answer.

(2)

- (b) By considering mixed strategies, find the ratio of the number of times Alan should play each of the other two shots.

(8)

- (c) Find the value of the game.

(1)

(Total 11 marks)

Q15.

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Alma and Brian play a zero-sum game. The game is represented by the following pay-off matrix for Alma.

		Brian			
		I	II	III	IV
Alma	I	3	-1	7	-7
	II	3	-6	0	8
	III	8	12	8	11
	IV	8	13	8	10

- (a) Write down the pay-off for **Brian** if Alma plays I and Brian plays IV.

(1)

- (b) Show that this game has a stable solution.

(3)

- (c) Find all the saddle points for the game.



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